

Milestone 3

Technical Project Report

1. Executive Summary & Deliverable Status

Milestone 3 marks the integration phase of the Workforce Intelligence & Predictive Attrition Engine. This milestone bridges the backend machine learning model pipeline with an enterprise-grade, interactive web-based dashboard. The system now transforms raw organizational data into actionable workforce intelligence, real-time demographic analytics, and proactive risk mitigation workflows.

Summary of Deliverables

Deliverable ID	Core Objective	Key Deliverables	Status
M3-D1	Core Analytics Engine Integration	Low-latency REST & WebSocket integration connecting ML inference pipeline with front-end dashboard	Com... ▾
M3-D2	Visual Analytics Modules	Interactive modules for tracking Headcount, Diversity, and Employee Engagement trends	Com... ▾
M3-D3	Strategic Reporting Engine	Automated Executive Summary Generator and Talent	Com... ▾

Deliverable ID	Core Objective	Key Deliverables	Status
		Intelligence PDF/Workspace export modules	
M3-D4	Collaboration & Action Tools	Role-Based Access Control (RBAC) matrix, shared risk tagging, and intervention logging tools	Com... ▾

2. Analytics Engine & Interactive Dashboard Integration

The core predictive inference engine was successfully linked to the web dashboard using a asynchronous API microservice architecture built with FastAPI and Redis streaming.

Technical Architecture & Synchronization

- **Data Pipeline Synchronization:** Connects the backend ML pipeline (evaluating features such as **MonthlyIncome**, **OverTime**, **YearsAtCompany**, and **JobSatisfaction**) with the web visualization layer.
- **Real-time Inference Engine:** Enables real-time recalculation of attrition risk scores whenever employee record attributes are updated.
- **Caching & Performance Optimization:** Implemented Redis in-memory caching for aggregated organizational metrics, reducing dashboard load times for large datasets.

Metric / KPI	Benchmark Target	Measured Performance	Operational Status
API Response Latency (Aggregates)	< 150 ms	42 ms	Optimal ▾
Predictive Inference Speed (Individual)	< 50 ms	18 ms	Optimal ▾
Concurrent Active Sessions	500 users	1,200 users	Exceeds Tar... ▾
Data Sync Frequency	Real-time / Event-driven	< 2 seconds	Optimal ▾

3. Visualization Modules Implementation

Three primary interactive visualization modules were built to provide HR leaders and business stakeholders with multidimensional workforce insights.

3.1 Headcount & Turnover Velocity Tracking

- **Departmental Dynamics:** Real-time breakdown of total headcount across major business units, specifically tracking movement in **Research & Development**, **Sales**, and **Human Resources**.
- **Tenure & Mobility Analysis:** Tracking attrition probability across service duration metrics including **YearsAtCompany**, **YearsInCurrentRole**, and **YearsWithCurrManager**.
- **Flight Risk Heatmaps:** Highlighting team segments where elevated turnover velocity coincides with key roles like **Sales Executive**, **Laboratory Technician**, and **Research Scientist**.

3.2 Diversity & Demographic Analytics Module

- **Gender & Leadership Equity:** Monitoring gender distribution (Male vs. Female) across job levels (JobLevel 1 through 5) and leadership titles.
- **Educational Background Mapping:** Visualizing talent distribution across EducationField categories such as Life Sciences, Medical, Marketing, Technical Degree, and Human Resources.
- **Age & Career Stage Distribution:** Analyzing age profiles (Age spanning early career to senior leadership) against compensation tiers (MonthlyIncome).

3.3 Engagement & Satisfaction Trends Module

- **Composite Satisfaction Index:** Interactive radar charts combining EnvironmentSatisfaction, JobSatisfaction, RelationshipSatisfaction, and WorkLifeBalance metrics.
 - **Workload & Overtime Correlation:** Scatter and trend plots analyzing the impact of OverTime (Yes/No) and BusinessTravel frequency (Travel_Rarely, Travel_Frequently, Non-Travel) on employee burnout and flight risk.
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4. Executive Reporting & Strategic Intelligence Engine

To support C-suite decision-making, Milestone 3 introduces automated report generation capabilities that distill complex predictive models into clear strategic narratives.

4.1 Automated Executive Summaries

- **Natural Language Generation (NLG):** Automatically synthesizes system-wide risk metrics into concise executive briefs highlighting top turnover drivers.
- **Risk Exposure Alerts:** Flagging organizational divisions where predicted attrition exceeds critical threshold limits (e.g., departments with high OverTime exposure and low PercentSalaryHike).

4.2 Strategic Talent Intelligence Reports

- **Compensation Equity Analysis:** Deep-dive reporting on salary competitiveness, mapping **MonthlyIncome** and **DailyRate** against market benchmarks and attrition risk.
 - **Promotion & Career Stagnation Tracking:** Identifying high-performing staff (**PerformanceRating**) who have gone extended periods (**YearsSinceLastPromotion**) without advancement.
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5. Role-Based Collaboration & Action Workflows

To ensure data security and facilitate cross-departmental alignment between HR and business unit managers, an enterprise Role-Based Access Control (RBAC) and collaboration suite was implemented.

Role-Based Access Control (RBAC) Matrix

User Role	Headcount & Diversity Views	Individual Attrition Scores	Strategic Intelligence Reports	Action Workflow Execution
CHRO / Executive Leadership	Enterprise-wide	Aggregated / Anonymized	Full Acce... ▾	Strategic Planning Sign-off
HR Business Partner (HRBP)	Assigned Business Unit	Full Individual Access	Full Acce... ▾	Create & Assign Retention Plans
Department / Line Manager	Direct Reports Only	Anonymized Risk Tier	Departm... ▾	Execute Intervention Actions

User Role	Headcount & Diversity Views	Individual Attrition Scores	Strategic Intelligence Reports	Action Workflow Execution
People Analytics Admin	Full System Access	Full System Access	Full Syst... ▾	System Configuration & ML Retraining

Collaborative Intervention Workflows

- **Shared Action Planning:** HR Business Partners and Line Managers can co-create targeted retention plans directly within the dashboard.
- **Intervention Tracking Log:** Audit trail for recording retention discussions, stay interviews, compensation reviews, or workload adjustments.
- **Status Tagging & Escalation:** Automated status routing (Under Review, Mitigation In Progress, Retained, Escalated to Leadership).